CHILLED WATER CIRCULATION SCHEMATIC & DIFFERENTIAL PRESSURE CONTROL:
1. Secondary & Tertiary chilled water circuit operating at 6.7°C supply / 13.3°C return temperature (Delta T = 6.6°C).
2. Variable primary and secondary pumping scheme utilizing end-suction centrifugal pumps with IE4 permanent magnet motors.
3. Differential pressure transmitter located at hydraulically most remote AHU coil modulating pump speed to maintain 1.8 bar DP.
4. Automatic equalized run-time rotation: System swaps duty and standby pumps every 168 operating hours.
5. High accuracy electromagnetic flow meter and paired PT1000 temperature sensors transmitting BTU cooling data to Central BMS.

Pump ID	Service Designation	Flow Rate	Head	Motor kW	Impeller Material
PMP-CHW-01	Primary CHW Pump (Duty 1)	350 m3/hr	22 Metres	37 kW IE4	SS 316 Stainless Steel
PMP-CHW-02	Primary CHW Pump (Duty 2)	350 m3/hr	22 Metres	37 kW IE4	SS 316 Stainless Steel
PMP-CHW-03	Primary CHW Pump (Standby)	350 m3/hr	22 Metres	37 kW IE4	SS 316 Stainless Steel
PMP-SEC-01	Secondary CHW Pump (Duty 1)	420 m3/hr	38 Metres	55 kW IE4	SS 316 Stainless Steel
PMP-SEC-02	Secondary CHW Pump (Duty 2)	420 m3/hr	38 Metres	55 kW IE4	SS 316 Stainless Steel
PMP-SEC-03	Secondary CHW Pump (Standby)	420 m3/hr	38 Metres	55 kW IE4	SS 316 Stainless Steel

AMARAVATI GOVERNMENT COMPLEX · LEGISLATIVE ASSEMBLY		
DWG TITLE: TERTIARY CHILLED WATER PUMP CONTROL PANEL SCH		
DRAWING NO: SL-288 _ AMARAVATI ASSEMBLY SC	REV: 1	SCALE: 1:100
CLIENT: AP CRDA / GOVT OF ANDHRA PRADESH	DATED: 28-09-2026	